

Mathematics Questions - Algebra

Subject: Mathematics | Chapter: Algebra | Difficulty: medium

Question 1: Solve the equation $3x^2 - 7x + 2 = 0$.

Options:

- A) $x = 2, x = 1/3$
- B) $x = -2, x = -1/3$
- C) $x = 2, x = -1/3$
- D) $x = -2, x = 1/3$

Difficulty: 3/5

Question 2: Simplify the expression $(2x^3y^2)(4x^1y^3)$.

Options:

- A) $8x^2y$
- B) $8x^2y^5$
- C) $8x^2y^6$
- D) $8x^4y^5$

Difficulty: 3/5

Question 3: Solve the system of equations: $2x + y = 7$ and $x - y = 2$.

Options:

- A) $x = 3, y = 1$
- B) $x = 1, y = 3$
- C) $x = 3, y = -1$
- D) $x = -1, y = 3$

Difficulty: 3/5

Question 4: What is the slope of the line passing through the points (4, 2) and (8, 6)?

Options:

- A) 1
- B) -1
- C) 2
- D) $\frac{1}{2}$

Difficulty: 2/5

Question 5: Expand and simplify $(x + 3)(x - 5)$.

Options:

- A) $x^2 - 2x - 15$
- B) $x^2 + 2x - 15$
- C) $x^2 - 8x - 15$
- D) $x^2 + 8x - 15$

Difficulty: 2/5

Question 6: If $f(x) = 2x + 1$, find $f^{-1}(x)$.

Options:

- A) $f^{-1}(x) = (x - 1)/2$
- B) $f^{-1}(x) = (x + 1)/2$
- C) $f^{-1}(x) = 2x - 1$
- D) $f^{-1}(x) = 2x + 1$

Difficulty: 4/5

Question 7: Solve the inequality $2x - 5 > 3$.

Options:

- A) $x > 4$
- B) $x < 4$
- C) $x > -4$
- D) $x < -4$

Difficulty: 2/5

Question 8: What are the x-intercepts of the quadratic function $y = x^2 - 4x + 3$?

Options:

- A) $x = 1, x = 3$
- B) $x = -1, x = -3$
- C) $x = 1, x = -3$
- D) $x = -1, x = 3$

Difficulty: 3/5

Question 9: Simplify the radical expression $\sqrt{75}$.

Options:

- A) $5\sqrt{3}$
- B) $3\sqrt{5}$
- C) $25\sqrt{3}$
- D) $3\sqrt{25}$

Difficulty: 2/5

Question 10: Solve for x: $\log_2(x) = 3$

Options:

A) $x = 6$

B) $x = 8$

C) $x = 9$

D) $x = 10$

Difficulty: 3/5

Generated on: 30/7/2025, 3:13:21 am